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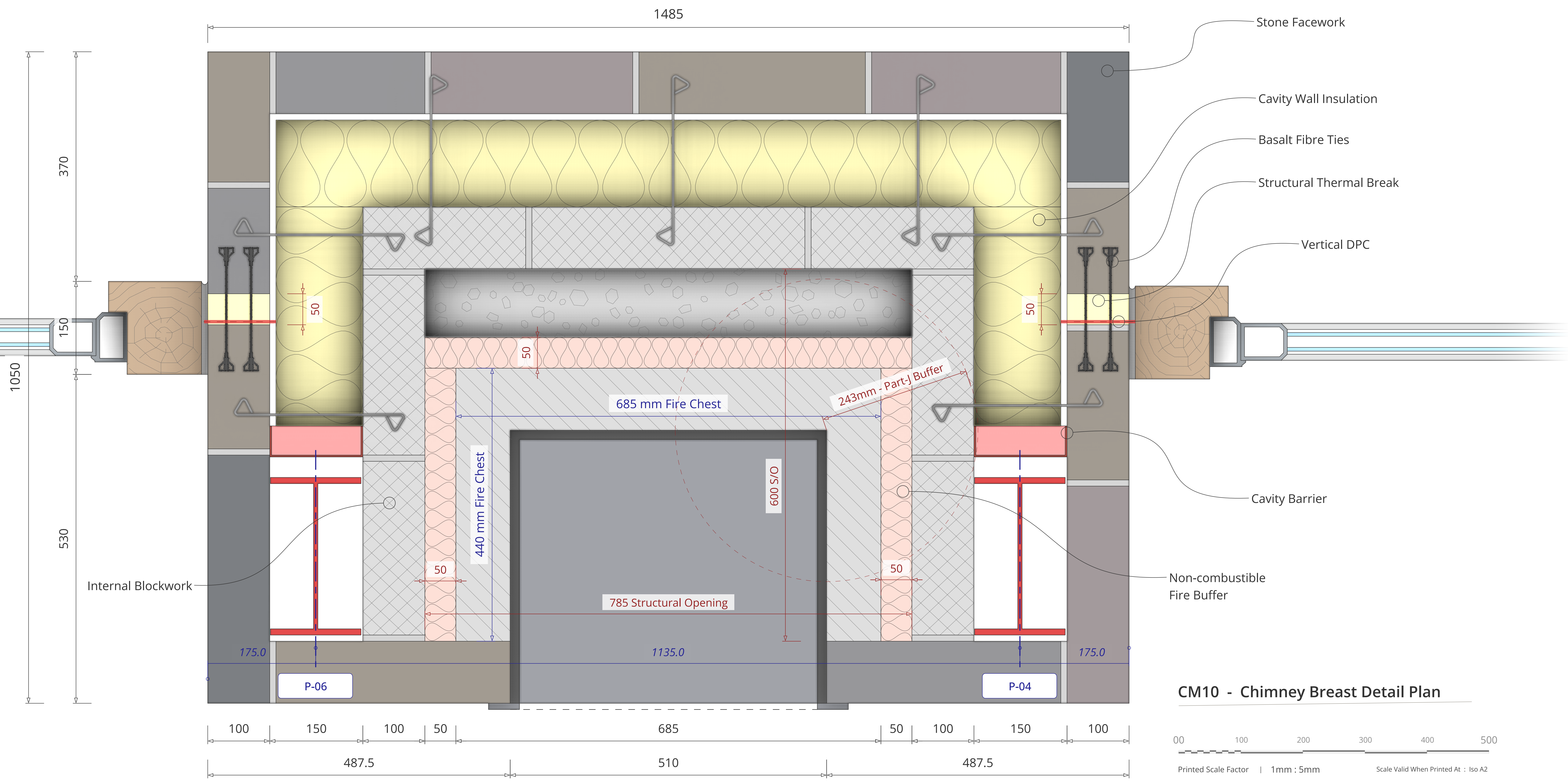

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.

CM10 - Chimney Breast Detail Plan

00 100 200 300 400 500

Printed Scale Factor | 1mm : 5mm

Scale Valid When Printed At : Iso A2

Element	Specifications
Stone Facework	Construct the outer leaf using 100mm Natural Stone facing. Bed the stone in a designation iii sand and cement mortar (approx 1:1:6 cement:lime:sand) incorporating hydrated lime or a quality mortar plasticiser to provide necessary flexibility against thermal expansion cracking while matching the adjacent brickwork aesthetic; avoid pure lime mortars to reduce maintenance repointing. Keep the cavity face clean as work proceeds.
Recommended Finish	The chimney envelope outer leaf is to be constructed using Blue Pennant Sandstone Random Facing Stone. This material is a hard-wearing, sedimentary gritstone quarried in South Wales, presenting a dominant palette of dark blue-greys and charcoal, interspersed with characteristic naturally occurring seams of coppery-orange and coal-grey. The stone is to be supplied 'random cropped', cut to varying heights and random lengths with mostly square edges to create a traditional, non-coursed random build aesthetic. The finish is a natural split-face texture that highlights the geological grain and rugged, rustic character of the material.
Cavity Insulation (Main Zone)	Install 140mm Recticel Eurowall PIR partial-fill cavity boards to the main cavity zones (Yellow Zone), strictly in accordance with specification WT10. Ensure boards are tightly butted and clips are used to maintain the residual cavity.
Cavity Insulation (Fire Buffer)	In the critical zone immediately surrounding the firechest (Pink Zone), install 50mm Rockwool Hardrock Multi-Fix (or A1 non-combustible equivalent). This 50mm thickness is sufficient to serve as the non-combustible heat buffer. CRITICAL SITE CHECK: While the side insulation is reduced to 50mm, the Bricklayer must verify that the distance from the firechest to the start of the 'Yellow' PIR still exceeds 200mm diagonally.

Element	Specifications
Inner Blockwork	Construct the inner structural core using 100mm Solid Dense Concrete Blocks (7N) to enclose the firechest and support the flue system. Ensure the blockwork courses align with the stone facing to facilitate the installation of the Ancon ties at every course (225mm). Keep the 50mm gap between this inner blockwork and the firechest clear of mortar droppings.
Structural Thermal Break	Secure the chimney structure to the main building leaf using Leviat Ancon Teplo-BF basalt fibre ties. Install these ties at 225mm vertical centres to ensure robust structural continuity without creating a thermal bridge. Ensure ties are fully bedded into fresh mortar on both sides of the insulation strip.
Cavity Barrier	CRITICAL FIRE STOP: Install a vertical continuous fire-rated cavity barrier (e.g., Rockwool Fire Barrier or Siderise) at the junction where the chimney cavity meets the main house wall cavity (aligned with the steel columns). This barrier must fill the full void width and provide minimum 30-minute fire integrity to prevent smoke or fire tracking from the chimney void into the main dwelling's combustible insulation zone.
Design Rationale	The design employs a thermally broken structural connection using Ancon Teplo ties to isolate the chimney interior and exterior thermal mass. A 'Hybrid' insulation strategy uses Recticel PIR for the main cavity envelope, and a second slimline 50mm Rockwool Euroclass A1 rated buffer zone near the appliance. This 50mm layer exceeds the manufacturer's minimum non-combustible clearance requirements of 30mm while ensuring full compliance with Approved Document J by ensuring the area with active flames is separated 200mm from the PIR filled high risk higher performance outer cavity wall insulation.